Lab exercise 8

Query with Sub Query& Correlated Query

AIM:

To perform subquery and correlated query on the given relations.

DESCRIPTION:

SUBQUERY

A MySQL subquery is a query nested within another query such as SELECT, INSERT, UPDATE or DELETE. In

addition, a MySQL subquery can be nested inside another subquery.

A MySQL subquery is called an inner query while the query that contains the subquery is called an outer query. A

subquery can be used anywhere that expression is used and must be closed in parentheses.

1. Which vegetable has a price greater than the average price?

Subquery:

```
SELECT VegName, Price
FROM Vegetable
WHERE Price > (SELECT AVG(Price) FROM Vegetable);
```

Output:

```
+----------+-------+
| VegName  | Price |
+----------+-------+
| Cabbage  | 60.00 |
| Beans    | 63.00 |
| Capsicum | 67.00 |
+----------+-------+
```

2. Which vegetables belong to a category having more than one vegetable?

Correlated Subquery:

```
SELECT V1.VegName, V1.Category
FROM Vegetable V1
WHERE (SELECT COUNT(*)
       FROM Vegetable V2
       WHERE V2.Category = V1.Category) > 1;
```

OUTPUT:

```
    ->          WHERE V2.Ca
+----------+-----------+
| VegName  | Category  |
+----------+-----------+
| Tomato   | Vegetable |
| Potato   | Vegetable |
| Onion    | Vegetable |
| Brinjal  | Vegetable |
| Cabbage  | Vegetable |
| Beans    | Vegetable |
| Capsicum | Vegetable |
+----------+-----------+
```

3. Which vegetable has the highest price?

Since your table does **not have a date/joined-date column**, the original faculty question cannot be directly applied.

A suitable **subquery** for your table is:

```
SELECT VegName, Price
FROM Vegetable
WHERE Price = (SELECT MAX(Price) FROM Vegetable);
```

VegName	Price
Capsicum	67.00

4. List the category and price of vegetables whose price is greater than the average price of their category.

Correlated Subquery:

```
SELECT V1.Category, V1.VegName, V1.Price
FROM Vegetable V1
WHERE V1.Price > (
    SELECT AVG(V2.Price)
    FROM Vegetable V2
    WHERE V2.Category = V1.Category
);
```

Category	VegName	Price
Vegetable	Cabbage	60.00
Vegetable	Beans	63.00
Vegetable	Capsicum	67.00